

HNSS15 · Head & Neck Pathology — Disease Chart

Oral cavity · Nose & upper airway · Larynx · Salivary gland

HoyaDocs

Updated Sep 29, 2026 · Rev 1

Oral Cavity & Oropharynx						
DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	PATHOLOGY	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Leukoplakia <i>the white warning</i> Premalignant (5–25%)	Adults 40–70; tobacco (smoked, chewed, snuff)	Tongue, floor of mouth, hard palate, buccal mucosa	White patch/plaque that cannot be scraped off and can't be called any other disease (diagnosis of exclusion)	Most show hyperkeratosis only at first biopsy; some show dysplasia (mild → severe)	—	Most in the US are benign; ~15% of dysplastic lesions → carcinoma within 5 yr. Biopsy if it won't scrape off
Hairy leukoplakia <i>the EBV impostor</i> Benign — NOT premalignant	Immunocompromised (HIV)	Lateral border of tongue	"Fluffy", corrugated white patches; can't be scraped off	Hyperkeratosis + balloon cells	EBV	Known cause, so not true leukoplakia; no cancer risk
Erythroplakia <i>the red flag</i> Almost always premalignant	Adults; less common than leukoplakia	Oral cavity (floor of mouth is high-risk)	Red, velvety flat patch	Thin, atypical epithelium: dysplasia, CIS or minimally invasive carcinoma	—	Much higher risk than leukoplakia (<i>white = warning, red = red flag</i>)
Keratinizing SCC (HPV-negative) <i>the smoker's cancer</i> Malignant	Middle-aged/older; tobacco + alcohol (multiply risk); betel quid/qaan; lower lip : sun + pipe smoking; rising in <40 with no risk factors (tongue)	Oral cavity (ventral tongue, floor of mouth, lower lip, soft palate), larynx	Non-healing ulcer , induration, pain (± referred ear pain), dysphagia	Ulcerated, irregular; keratin pearls/whorls , invasion. Path: hyperkeratosis → dysplasia → CIS → invasion	TP53, CDKN2A (p16 lost), PIK3CA, NOTCH1, FAT1; more mutations	Field cancerization → second primaries. Spread: cervical nodes → lung, bone. 5-yr: ~80% early, ~20% late. Stage, not grade
HPV-positive SCC <i>the young non-smoker's tonsil cancer</i> Malignant — better prognosis	Often younger, non-smoker; HPV-16 (~90%)	Oropharynx: tonsils and base of tongue	Often a painless neck mass (cystic node) first	Non-keratinizing : nests of basaloid cells	E6 → p53, E7 → Rb → p16 overexpression (stain nucleus + cytoplasm); HPV mRNA by ISH	Better survival than classic SCC

★ **USMLE high-yield:** White patch that won't scrape off = leukoplakia → biopsy (scrapes off = Candida). Red velvety patch = erythroplakia, almost always dysplasia/CIS. Tonsil or base of tongue in a young non-smoker = HPV-16, p16+. In HNSCC, **stage, not grade**, predicts outcome.

Nose & Upper Airway

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	PATHOLOGY	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Pyogenic granuloma <i>the misnamed bleeder</i> Benign	Any age; pregnancy ; trauma/hormones suspected	Gingiva , skin, any mucosa including nose	Red, raised papule ; bleeds easily, ulcerates → epistaxis	Lobular capillary hemangioma ; neither infectious nor granulomatous	—	Excision or sclerosing injection
Juvenile nasopharyngeal angiofibroma <i>the teenage boy's bleeding mass</i> Benign, locally aggressive	Adolescent boys (7–19; rare >25); sporadic or FAP ; hormonal theory	Nasopharynx; near posterior attachment of middle turbinate	Obstruction (80–90%), epistaxis (45–60%), headache, facial swelling, one-sided rhinorrhea; fills one side, septum pushed to the other	Dense collagen , bland fibrosis, thin-walled vessels	CTNNB1 (β-catenin) — same Wnt pathway as APC in FAP	Intracranial extension or young age = worse. Surgery
Sinonasal (Schneiderian) papilloma <i>one-sided obstruction in a middle-aged man</i> Benign; some types turn malignant	Men 30–60	Nasal cavity/paranasal sinus; almost always unilateral	Nasal obstruction; coexisting sinusitis (tumor blocks drainage)	Columnar or squamous epithelium + mucous cysts . Types: exophytic, inverted (grows into stroma), oncocytic	EGFR (inverted type). HPV found but not proven causal	Malignant risk: Exophytic ~0 < Inverted 5–10% < Oncocytic 14–19% (→ SCC). Complete excision
Nasopharyngeal carcinoma <i>the EBV cancer</i> Malignant	Geographic: southern China/SE Asia adults , African children ; EBV	Lateral wall of nasopharynx	Obstruction + epistaxis; neck mass (nodal mets); one-sided serous otitis media (blocks eustachian tube); often diagnosed late	Keratinizing (lecture: most common), non-keratinizing (small dark cells with lymphocytes), basaloid	EBV (EBER by in situ hybridization)	Non-keratinizing does better (radiosensitive)

★ **USMLE high-yield:** Teenage boy + one-sided obstruction + nosebleeds = angiofibroma (β-catenin, FAP). Adult with one-sided middle-ear fluid + neck node + Asian background = nasopharyngeal carcinoma (EBV). Pyogenic granuloma is a capillary hemangioma, not an infection or granuloma.

Larynx

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	PATHOLOGY	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Laryngeal papilloma / juvenile papillomatosis <i>the recurring childhood wart</i> Benign	Children (multiple) or adults (solitary); HPV 6/11 from mother during birth	True vocal cords ; can spread down the airway	Hoarseness , breathing difficulty; may bleed	Small (<1 cm), exophytic; finger-like projections of orderly squamous epithelium over fibrovascular cores	Low-risk HPV 6/11	Children: recur often → many surgeries; regress at puberty . Airway obstruction; lecture: rare SCC change (poor prognosis). Surgery; HPV vaccine prevents
Laryngeal SCC <i>the smoker's hoarse voice</i> Malignant	Men, 60s–70s; smoking (+ alcohol)	Vocal cords, epiglottis, aryepiglottic folds, pyriform sinuses	Persistent hoarseness or dysphagia	Plaque-like or ulcerated, fungating ; variable differentiation; arises from dysplasia (high-grade = highest risk)	—	Stage predicts prognosis. Glottic tumors found early, few lymphatics → better. Early: minimally invasive; late: surgery + chemo + radiation

★ **USMLE high-yield:** Child + hoarseness + recurring vocal cord masses = juvenile papillomatosis (HPV 6/11, acquired at birth) — they **recur**. Older male smoker + persistent hoarseness = laryngeal SCC. Don't confuse with **vocal cord nodules**: bilateral, from voice overuse, not neoplastic.

Salivary Gland (Parotid)

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	PATHOLOGY	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Pleomorphic adenoma <i>the most common one — and it recurs</i> Benign · #1 overall	Adults; radiation exposure	Parotid (below/in front of ear)	Slow-growing, painless, mobile mass	Benign mixed tumor: epithelial (ductal) + myoepithelial + mesenchymal; chondromyxoid stroma; well circumscribed ± capsule	PLAG1 overexpression	Recur s if not completely excised; rare malignant change (carcinoma ex pleomorphic adenoma)
Warthin tumor <i>the smoker's parotid tumor</i> Benign · #2 overall	Adult men > women ; smoking ; ~10% bilateral	Parotid only	Painless mass	Papillary cystadenoma lymphomatosum: cysts with a double layer of oncocytic cells (columnar + cuboidal) over lymphoid tissue with germinal centers	Oncocytes packed with mitochondria	Low recurrence, good prognosis
Adenoid cystic carcinoma <i>the painful, nerve-invading one</i> Malignant	Adults	Minor salivary glands (~50%) + parotid; also lung, breast, sinuses	Slow-growing mass with pain , numbness, facial palsy	Small, dark, round cells; cribriform ("Swiss cheese"); poorly encapsulated; perineural invasion	MYB::NFIB → activates NOTCH1	Unpredictable; recur s stubbornly , spreads years later
Mucoepidermoid carcinoma <i>the most common malignant one</i> Malignant · #1 malignant	Adults and children	Parotid (any salivary gland)	Mass; can grow large	Gray-white, mucinous , partly cystic; mix of mucus + intermediate + squamous cells; infiltrative	t(11;19) → CRTC1::MAML2 (~half)	Prognosis by GRADE : low grade recurs, rarely spreads; high grade spreads, ~50% 5-yr survival

★ **USMLE high-yield:** Most salivary tumors are in the parotid and most parotid tumors are benign — but **the smaller the gland, the more likely malignant** (submandibular ~40%, minor ~50%, sublingual 70–90%). **Pain or facial nerve palsy = malignant**. Mucoepidermoid: **grade** matters (the opposite of HNSCC).

HNSS16 · Head & Neck Pathology Lab — Disease Chart

Oral cavity · Nose & upper airway · Larynx · Salivary glands · Benign neck cysts | **NEW in HNSS16** tags mark diseases not on the HNSS15 chart

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Oral Cavity & Oropharynx						
DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	GROSS & MICRO (LAB ID)	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Oral candidiasis (thrush) <i>the white patch that wipes off</i> Infectious NEW in HNSS16	Immunocompromised, neutropenia ; antibiotics, inhaled steroids	Tongue, oral mucosa	White, slightly raised patch that scrapes off with a tongue blade, leaving a red (erythematous) base	Yeast + pseudohyphae , fibrin, inflammatory cells	<i>Candida</i>	Treat the infection; the key distinction from leukoplakia is that it wipes off
Leukoplakia <i>the white warning</i> Premalignant (5–25%)	Adults; tobacco	Any oral mucosa	White lesion that cannot be scraped off ; diagnosis of exclusion (not infection or trauma)	Hyperkeratosis ± dysplasia : loss of maturation, nuclear pleomorphism (lab image shows severe dysplasia)	—	Only 5–25% premalignant. Biopsy to look for dysplasia
Erythroplakia <i>the red flag</i> More directly premalignant	Men; long-standing tobacco (look for tobacco-stained teeth)	Oral mucosa	Red , velvety patch	Dysplasia, CIS or early invasive carcinoma	—	More directly premalignant than leukoplakia
HPV-associated SCC <i>the young adult's small tumor with a big neck node</i> Malignant — good prognosis	Younger adults; number of oral sex partners ; high-risk HPV-16	Oropharynx (tonsils, base of tongue)	Small solitary primary with an enlarged cervical lymph node	Non-keratinizing ; nests of basaloid cells	HPV-16; p16+ (E7 → Rb)	Good outcome; distant mets rare; low risk of second primary
Tobacco/alcohol SCC <i>the older smoker's bulky tumor</i> Malignant — poor prognosis	Older ; tobacco + alcohol (each a risk; combined risk multiplies)	Oral cavity ; lab gross shows base of tongue	Large, bulky primary ± nodes	Keratinizing : prominent keratin pearls	TP53, CDKN2A (p16 lost)	Poor outcome; distant mets common; multiple primaries (field cancerization)

HPV vs. non-HPV SCC — learn this table

	AGE	RISK FACTORS	LOCATION	PRESENTATION	HISTOLOGY	DISTANT METS	OUTCOME	2ND PRIMARY
HPV-associated	Younger	No. of oral sex partners (HPV-16)	Oropharynx	Small primary + bulky nodes	Non-keratinizing (p16+)	Rare	Good	Low
Non-HPV	Older	Tobacco + alcohol	Oral cavity	Large primary , variable nodes	Keratinizing (pearls)	Common	Poor	High

★ **USMLE high-yield:** White + **wipes off** = Candida (yeast + pseudohyphae). White + **won't** wipe off = leukoplakia → biopsy. Red = erythroplakia, more directly premalignant. **Keratin pearls** = **tobacco/alcohol SCC**; non-keratinizing + small primary + big node in a young adult = HPV-16. Tongue terms: **macroglossia** = big tongue; **atrophic glossitis** = smooth atrophic tongue from anemia/vitamin deficiency.

Nose & Upper Airway

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	GROSS & MICRO (LAB ID)	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Juvenile nasopharyngeal angiofibroma <i>the teenage boy's bleeding mass</i> Benign, locally aggressive	Adolescent boys; associated with FAP (often co-occur)	Fibrovascular stroma of the posterolateral wall and roof of the nasal cavity	Frequent epistaxis ; also obstruction	"Angio" : blood vessels with RBCs + "fibroma" : dense collagenous stroma; no mitotic activity	β-catenin (CTNNB1) — same Wnt pathway as FAP	Surgery; recurrence is common
Sinonasal (Schneiderian) papilloma <i>one-sided obstruction in a middle-aged man</i> Benign; some types turn malignant	Adult men; some HPV-positive	Nasal cavity or paranasal sinuses; inverted type on the lateral nasal wall	Nasal obstruction	Inverted : squamous epithelium growing inward into the stroma . Exophytic: outward papillary growth. Oncocytic (cylindrical cell)	EGFR (inverted type)	Exophytic: low risk. Inverted: malignant potential (→ SCC). Oncocytic: may recur. See <i>the heads-up box on which type is highest risk</i> .
Nasopharyngeal carcinoma <i>the EBV cancer</i> Malignant	Southeast Asia / southern China ; EBV	Nasopharynx	Obstruction ± epistaxis; neck node	Large clusters of epithelial cells surrounded by lymphocytes . Keratinizing and non-keratinizing subtypes	EBV → non-keratinizing subtype	Non-keratinizing = better prognosis (radiosensitive)

Compare by presenting symptom (LO)

MAINLY PRESENTS AS...	LESIONS
Epistaxis (nosebleeds)	Juvenile nasopharyngeal angiofibroma (frequent bleeding) · pyogenic granuloma
Obstruction	Sinonasal papilloma · nasopharyngeal carcinoma (± bleeding) · angiofibroma (also obstructs)
Hoarseness	Singer's nodule · laryngeal polyp · laryngeal papilloma · laryngeal SCC

★ **USMLE high-yield**: Teenage boy + nosebleeds + **vessels in dense collagen, no mitoses** = angiofibroma (β-catenin, FAP). Epithelium **growing inward** into stroma = inverted sinonasal papilloma (EGFR). Epithelial clusters **within lymphocytes** + Asian patient = nasopharyngeal carcinoma (EBV, non-keratinizing).

⚠ **Heads-up**: HNSS15 ranks **oncocytic** papilloma highest for malignancy (14–19% vs. inverted 5–10%). The HNSS16 lab review calls **inverted** the highest. If asked, inverted is the type most tied to malignancy and EGFR in both; know that oncocytic also carries real risk.

Larynx

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	GROSS & MICRO (LAB ID)	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Laryngeal (singer's) nodule <i>the overuser's bilateral bumps</i> Benign, reactive NEW in HNSS16	Voice overuse (singers, teachers)	Vocal folds, bilateral	Hoarseness	Hyperplastic surface epithelium over edematous submucosa	—	Low risk of carcinoma
Laryngeal polyp <i>the smoker's one-sided polyp</i> Benign NEW in HNSS16	Older patients who smoke	Vocal fold, unilateral	Hoarseness	Similar to a nodule (differential)	—	Benign; key difference from nodule is one-sided + smoker
Laryngeal papilloma / papillomatosis <i>the recurring childhood wart</i> Benign	Adults: solitary . Children: multiple , HPV acquired through the birth canal	True vocal cord	Hoarseness ; bleeding from surface ulceration	Papillary projections of squamous epithelium around fibrovascular cores	HPV (6/11)	Can progress to SCC, especially in younger patients. HPV vaccine prevents it
Laryngeal SCC <i>the smoker's hoarse voice</i> Malignant	Chronic smokers	Larynx (lab gross: laryngectomy specimen)	Hoarseness, cough, dysphagia ; bulky lesion	Usually well-differentiated, keratinizing SCC ; malignant cells invading past the basement membrane	—	Prognosis by stage

★ **USMLE high-yield:** Hoarseness differential: **bilateral + voice overuse** = singer's nodule (edematous submucosa). **Unilateral + older smoker** = polyp. **Papillary fronds with fibrovascular cores**, multiple in a child = papillomatosis (HPV from the birth canal). Bulky mass + keratinization + invasion in a chronic smoker = SCC.

Benign Neck Cysts

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	GROSS & MICRO (LAB ID)	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Branchial cleft cyst <i>the lateral neck lump</i> Benign NEW in HNSS16	Children/young adults (congenital)	Lateral neck (anterior to SCM)	Painless lateral neck mass; may enlarge with infection	Squamous-lined cyst with lymphoid tissue in the wall	Remnant of the branchial (pharyngeal) apparatus (2nd cleft most common)	Excision
Thyroglossal duct cyst <i>the midline lump that moves with swallowing</i> Benign NEW in HNSS16	Children/young adults (congenital)	Midline neck, near the hyoid	Midline mass that moves up with swallowing or tongue protrusion	Often lined by respiratory epithelium ; thyroid follicles in the wall	Remnant of the thyroglossal duct (thyroid descent from foramen cecum)	Excision (Sistrunk procedure, including mid-hyoid)

★ **USMLE high-yield:** Location decides it: **lateral** = branchial cleft cyst (squamous lining + **lymphoid tissue**). **Midline** = thyroglossal duct cyst (respiratory lining + **thyroid follicles**; moves with swallowing).

Salivary Glands

DISEASE	WHO / RISK FACTORS	WHERE	PRESENTATION	GROSS & MICRO (LAB ID)	MOLECULAR / MARKER	BEHAVIOR & PROGNOSIS
Acute sialadenitis <i>the stone-blocked gland</i> Inflammatory NEW in HNSS16	Duct obstruction by a stone (sialolithiasis) ; infection	Salivary gland	Painful, swollen gland	Pink-red acute inflammatory infiltrate (neutrophils)	—	Resolves when the obstruction/infection is treated
Chronic sialadenitis <i>the scarred, smoldering gland</i> Inflammatory NEW in HNSS16	Repeated obstruction, infection or autoimmunity	Salivary gland	Recurrent swelling	Lymphoid aggregates + bands of fibrosis (hallmarks of chronic inflammation anywhere)	—	Gland scarring
Sjögren syndrome (lymphoepithelial lesion) <i>the dry-eyed, dry-mouthed autoimmune gland</i> Autoimmune NEW in HNSS16	Autoimmune; classically middle-aged women	Bilateral parotids ; lacrimal glands	Dry mouth + dry eyes ; bilateral parotid enlargement	Dense lymphoid infiltrate with islands of residual epithelial and myoepithelial cells	—	Chronic; ↑ risk of lymphoma (Step 1)
Pleomorphic adenoma <i>the most common one — and it recurs</i> Benign	Adults	Parotid	Painless mass	Bluish-white (chondromyxoid) stroma with clusters of epithelial cells; normal gland at the edge	PLAG1	Recurr s if any tissue is left → resected with a wide margin
Warthin tumor <i>the smoker's parotid tumor</i> Benign	Smokers ; can be bilateral	Parotid	Painless mass	Papillary structures covered by eosinophilic (oncocytic) epithelium over lymphoid tissue with germinal centers	Oncocytes = mitochondria	Benign; good prognosis
Mucoepidermoid carcinoma <i>the most common malignant one</i> Malignant	Adults and children	Major salivary glands	Mass	Cords and sheets of 3 cell types: squamous, mucous, intermediate	t(11;19) → fusion protein disrupting Notch signaling	Prognosis by grade
Adenoid cystic carcinoma <i>the painful, nerve-invading one</i> Malignant	Adults	Most common malignancy of minor salivary glands	Slow-growing, painful (invades nerves)	Cribriform glands (“punched-out” holes) surrounded by hyaline basement-membrane-like material	MYB::NFIB	“Stubbornly recurrent” : hard to fully excise from minor gland sites

★ **USMLE high-yield:** Parotid = largest gland and most common tumor site, but tumors there are **more likely benign**. **Lymphoid aggregates + fibrosis** = chronic sialadenitis. **Bilateral parotid swelling + dry eyes/mouth** = Sjögren. Lab IDs: **chondromyxoid stroma** (pleomorphic) · **oncocytes + germinal centers** (Warthin) · **3 cell types** (mucoepidermoid) · **cribriform + hyaline** (adenoid cystic).